XRPL Validator — Daily Engineering Report · 2026-08-02

Period: 2026-08-02 00:00 to 24:00 US Eastern (2026-08-02 04:00 to 2026-08-03 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): remove self-crossed offers off the book tip** (`b733ae7`, xrpl-validator) rippled DELETES the taker's own offers off the tip rather than crossing them — BookStep.cpp:415-455 `limitSelfCrossQuality`, which explains the choice itself: "We could skip over the self offer in the book and only cross offers that are not our own. This would make a lot of sense, but we don't do it ... we can only operate on the tip of the order book. We can't leave an offer behind -- it would sit on the tip and …
- **fix(ledger): measure a payment hop's carry, don't subtract a sentinel** (`7efe2fd`, xrpl-validator) An intermediate hop of a multi-hop payment has no real ceiling, so it was handed a sentinel want-cap of (9.99e15, 60) and its output recovered as `want_cap - remaining`. That cannot work. `me_rescale` saturates (saturating_mul), so rescaling the sentinel down to a fill's exponent pins it at u128::MAX — and it does so again on the NEXT fill, resetting the running remainder and ERASING every fill before it. The …
- **feat(ledger): AMMDeposit tfTwoAsset equalDepositLimit + tecAMM_FAILED** (`a533219`, xrpl-validator) Amount and Amount2 on a two-asset deposit are MAXIMA, not the amounts deposited. rippled derives one side from the other at the pool's ratio and fails the transaction outright when NEITHER derivation fits inside what was offered (AMMDeposit.cpp:721-787): frac = amount / amountBalance (Number, 16 digits) tokensAdj = adjustLPTokens(lpt, multiply(lpt, frac, DOWNWARD)) frac = tokensAdj / lpt (adjustFracByTokens) amt2Dep …
- **refactor(ledger): one strand code path in Payment** (`c858f1f`, xrpl-validator) The payment had two execution shapes — a hop chain when Paths named usable intermediate currencies, and a separate direct cross_engine_to otherwise. They were the same walk: with no hops the chain is [spend, want], the loop runs exactly one hop with last=true, benef=dest and want_cap=want_gross, and issues the identical cross_engine_to call the direct branch made. The inflight snapshot iterates `hops` and so is …
- **feat(ledger): flow the default path alongside Paths, per pass** (`99f8a7a`, xrpl-validator) rippled ALWAYS builds the default path beside the ones named in Paths (unless tfNoRippleDirect), flows each strand, and applies the best-quality PASS each round before re-evaluating (StrandFlow.h:682-805). We ran the specified chain alone.

Independence program

No independence record for UTC day 2026-08-02; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle-Gated Stop/Trigger Offers** (dex) Add a tfStopOrder flag to OfferCreate plus SFields sfTriggerOracleID (a PriceOracle keylet), sfTriggerPrice, and sfTriggerDirection; a flagged offer i
- **Auditor-Keyed Ciphertext for Permissioned MPT Flows** (audit) Add a new owner-reserved ledger object AuditorKey (ltAUDITOR_KEY, keylet::auditorKey(account)) set by a new AuditorKeySet transaction carrying sfAudit
- **Sponsor-Backed Trust Line Reserves** (fees) Add a new owner-reserved ledger object SponsorAuthorization (ltSPONSOR_AUTH, keylet::sponsorAuth(sponsor, domain)) created by a new SponsorAuthorize t

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-02 04:49	—	xrpl-validator	`b733ae7`	fix(ledger): remove self-crossed offers off the book tip
08-02 15:41	—	xrpl-validator	`7efe2fd`	fix(ledger): measure a payment hop's carry, don't subtract a sentinel

Committed	Merged into main	Repo	Commit	Subject
08-03 01:56	—	xrpl-validator	`a533219`	feat(ledger): AMMDeposit tfTwoAsset equalDepositLimit + tecAMM_FAILED
08-03 02:41	—	xrpl-validator	`c858f1f`	refactor(ledger): one strand code path in Payment
08-03 03:30	—	xrpl-validator	`99f8a7a`	feat(ledger): flow the default path alongside Paths, per pass

Times in UTC. A commit appears on the day it was committed to main or merged into it.